

FACTSHEET FS00 · Overview and gap analysis

What changed on 2 October 2026. Two corrections, both to the universe of stocks, and every number below is re-run on them. (1) **US:** the universe is now the actual S&P 500 members at each month-end. Until now it was the 500 largest of every company that was *ever* in the index 2012–2026, which includes later winners before they joined. (2) **UK and EU:** the price panels now include the index members that stopped trading (UK coverage of members 71% → 92%, EU 81% → 89%); before, most of them had no prices. SCANDI stays as registered (OMXC25 + OMXS30 + OMXH25). Registered verdicts are unchanged; where a corrected number crosses a gate, the text says so. Comparison and method: FACTSHEETS_RERUN_2026-10-02.md.

US, 1999–2012 (pre-registered second period, run 2 October 2026). The same rules on Sharadar's point-in-time S&P 500 with delisted stocks, 1999–2012, a period these factsheets were not built on. 65 trials. What holds: the Turtle loss (FS01) and the low-volatility long-only alpha (FS04). What does not: momentum in every form (FS03, FS11, FS13c), and the 2013–2026 US losses for seasonality and the 52-week low. The checklist (FS20) passes in 1999–2012 only. The live book's US leg does not replicate (alpha t +1.10, needed 2). The trial ledger now holds 772 trials (Bonferroni $|t| > 3.99$). Pre-registration, method and every result: PREREG_US_1999_2012.md.

Twelve famous strategies, six markets, one set of rules: what we know, and what is missing

Henrik Gjerner · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · data to 31 August 2026

In one paragraph. Twelve strategies were run with the same code on six point-in-time universes (US, EU, UK, Denmark, Scandinavia, World), 2013–2026, after costs, each with a full factsheet: P&L, trade records, drawdowns, attribution, a pre-declared fix ladder, a classification and a 360° view. Only 0 of eleven long/short books (none) earn a **positive** return that passes the 2.87 gate in at least one market; 4 (FS01, FS06, FS08, FS12) pass it with a significantly **negative** return; long-only books beat the equal-weight universe's Sharpe most often in the momentum and low-risk families. Many signals are **redundant**: once their nearest library neighbours are in the model, 4 of eleven have no alpha left. The price-only battery collapses into a few roots: **momentum** (12-1, 52-week high, residual momentum), **low risk** (volatility, beta, MAX) and, weakly, **reversal** and **size**. The biggest gaps are data (fundamentals and delisted prices outside the US), models (no multivariate test yet) and implementation (capacity outside the US). Borrow fees, financing of leverage, World in one currency, a replication check, a programme-wide trial ledger and a market-impact cost model were added on 27 September 2026 (§3.0).

1. All strategies in one table

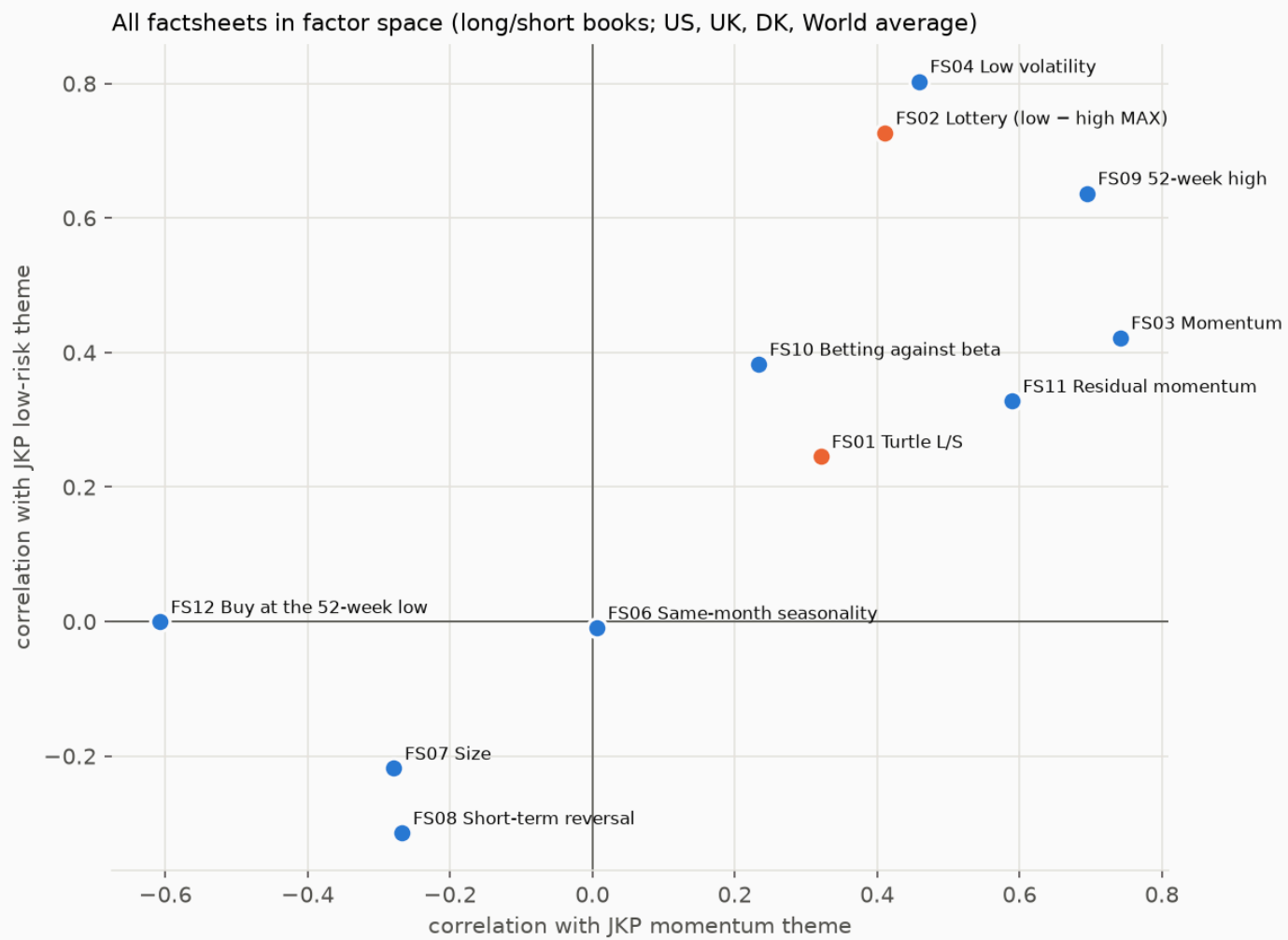
Strategy	Family	L/S net return / yr (range over 6)	L/S positive	L/S passes t > 2.87	L/S CAPM alpha range	Long-only beats EW Sharpe	Long-only alpha passes	Fixes: holdout Sharpe	Redundant in 360° view	Library cluster
FS01 Turtle Traders	Breakout & trend	-16.2% to -3.5%	0 / 6	US–, SCANDI–, World–	-12.3% to +2.7%	0 / 6	—	T0 -0.62 → T3 0.50 (drop shorts, 4N stop)	yes	momentum (time series)
FS02 Lottery (MAX)	Lottery & attention	-10.6% to +1.9%	1 / 6	—	-2.5% to +8.5%	2 / 6	—	L0 -0.41 → L1 -0.21 (beta-neutral legs)	yes	low risk
FS03 Momentum	Momentum & trend	-2.1% to +9.8%	5 / 6	—	+2.1% to +16.7%	5 / 6	UK, World	0.18 → 0.56 (full ladder)	no	momentum
FS04 Low volatility	Low risk	-8.7% to +1.6%	1 / 6	—	+2.8% to +10.9%	6 / 6	—	-0.14 → 0.21 (full ladder)	no	low risk
FS05 Monkey portfolios	Portfolio folklore	beat the index: 3–83% of monkeys	—	—	—	—	—	no fix: a benchmark	—	size (equal weight)
FS06 Same-month seasonality	Calendar & seasonality	-14.0% to +3.4%	1 / 6	EU–, DK–	-13.5% to +2.0%	1 / 6	—	-0.89 → -0.68 (full ladder)	no	momentum
FS07 Size	Size	-4.3% to +4.3%	1 / 6	—	-10.4% to +3.1%	1 / 6	UK	-0.34 → -0.37 (full ladder)	yes	low risk
FS08 Short-term reversal	Reversal & bottom-fishing	-12.4% to +0.7%	1 / 6	EU–	-15.7% to -0.7%	0 / 6	—	-0.87 → -0.98 (full ladder)	no	momentum
FS09 52-week high	Momentum & trend	-5.8% to +8.8%	2 / 6	—	+6.3% to +18.6%	5 / 6	EU, DK	0.04 → 0.52 (full ladder)	no	momentum
FS10 Betting against beta	Low risk	+1.7% to +9.1%	6 / 6	—	+1.1% to +7.7%	6 / 6	UK	0.01 → 0.06 (full ladder)	no	low risk
FS11 Residual momentum	Momentum & trend	+2.2% to +6.9%	6 / 6	—	+4.7% to +9.8%	6 / 6	—	0.50 → 0.58 (full ladder)	no	momentum
FS12 Buy at the 52-week low	Reversal & bottom-fishing	-14.7% to -5.8%	0 / 6	US–, EU–, World–	-15.8% to -7.3%	0 / 6	—	-0.78 → -0.57 (full ladder)	yes	momentum

Strategy	Family	L/S net return / yr (range over 6)	L/S positive	L/S passes t > 2.87	L/S CAPM alpha range	Long-only beats EW Sharpe	Long-only alpha passes	Fixes: holdout Sharpe	Redundant in 360° view	Library cluster
FS01 Turtle Traders	Breakout & trend	-16.2% to -3.5%	0 / 6	US–, SCANDI–, World–	-12.3% to +2.7%	0 / 6	—	T0 -0.62 → T3 0.50 (drop shorts, 4N stop)	yes	momentum (time series)
FS02 Lottery (MAX)	Lottery & attention	-10.6% to +1.9%	1 / 6	—	-2.5% to +8.5%	2 / 6	—	L0 -0.41 → L1 -0.21 (beta-neutral legs)	yes	low risk
FS03 Momentum	Momentum & trend	-2.1% to +9.8%	5 / 6	—	+2.1% to +16.7%	5 / 6	UK, World	0.18 → 0.56 (full ladder)	no	momentum
FS04 Low volatility	Low risk	-8.7% to +1.6%	1 / 6	—	+2.8% to +10.9%	6 / 6	—	-0.14 → 0.21 (full ladder)	no	low risk
FS05 Monkey portfolios	Portfolio folklore	beat the index: 3–83% of monkeys	—	—	—	—	—	no fix: a benchmark	—	size (equal weight)
FS06 Same-month seasonality	Calendar & seasonality	-14.0% to +3.4%	1 / 6	EU–, DK–	-13.5% to +2.0%	1 / 6	—	-0.89 → -0.68 (full ladder)	no	momentum
FS07 Size	Size	-4.3% to +4.3%	1 / 6	—	-10.4% to +3.1%	1 / 6	UK	-0.34 → -0.37 (full ladder)	yes	low risk
FS08 Short-term reversal	Reversal & bottom-fishing	-12.4% to +0.7%	1 / 6	EU–	-15.7% to -0.7%	0 / 6	—	-0.87 → -0.98 (full ladder)	no	momentum
FS09 52-week high	Momentum & trend	-5.8% to +8.8%	2 / 6	—	+6.3% to +18.6%	5 / 6	EU, DK	0.04 → 0.52 (full ladder)	no	momentum
FS10 Betting against beta	Low risk	+1.7% to +9.1%	6 / 6	—	+1.1% to +7.7%	6 / 6	UK	0.01 → 0.06 (full ladder)	no	low risk
FS11 Residual momentum	Momentum & trend	+2.2% to +6.9%	6 / 6	—	+4.7% to +9.8%	6 / 6	—	0.50 → 0.58 (full ladder)	no	momentum
FS12 Buy at the 52-week low	Reversal & bottom-fishing	-14.7% to -5.8%	0 / 6	US–, EU–, World–	-15.8% to -7.3%	0 / 6	—	-0.78 → -0.57 (full ladder)	yes	momentum

Net of costs, local currency, Feb 2013 – Aug 2026 (FS01 daily from Jan 2013). In the gate column + marks a significantly positive return, – a significantly negative one. The gate is Bonferroni over 12 cells per factsheet; fix ladders are judged on the 2020–26 holdout. Redundant = $\alpha |t| < 2$ in every universe after the signal's closest library neighbours and the market.

Eleven strategies, six markets, 2013-2026 (FS05 monkeys are shown separately)

Long/short Sharpe ratio (net)							Long-only Sharpe minus equal-weight universe Sharpe						
	US	EU	UK	DK	SCANDI	World		US	EU	UK	DK	SCANDI	World
FS01 Turtle Traders	-0.91	-0.49	-0.17	-0.35	-0.88	-0.75	FS01 Turtle Traders	-0.57	-0.25	-0.13	-0.74	-0.70	-0.46
FS02 Lottery (MAX)	-0.60	+0.13	-0.30	-0.22	-0.13	-0.44	FS02 Lottery (MAX)	-0.07	+0.14	+0.14	-0.16	-0.14	-0.05
FS03 Momentum	+0.08	+0.49	+0.41	+0.34	-0.14	+0.31	FS03 Momentum	+0.03	+0.21	+0.45	+0.24	-0.13	+0.24
FS04 Low volatility	-0.34	+0.08	-0.28	-0.06	-0.19	-0.27	FS04 Low volatility	+0.04	+0.32	+0.25	+0.24	+0.24	+0.15
FS06 Same-month seasonality	-0.70	-0.76	+0.24	-0.94	-0.63	-0.27	FS06 Same-month seasonality	-0.36	-0.39	+0.08	-0.45	-0.24	-0.12
FS07 Size	-0.13	-0.37	+0.43	-0.06	-0.01	-0.10	FS07 Size	-0.32	-0.33	+0.39	-0.12	-0.18	-0.13
FS08 Short-term reversal	-0.31	-0.79	-0.32	-0.42	+0.05	-0.50	FS08 Short-term reversal	-0.43	-0.56	-0.30	-0.27	-0.14	-0.38
FS09 52-week high	-0.24	+0.42	-0.04	+0.36	-0.07	-0.09	FS09 52-week high	-0.15	+0.51	+0.22	+0.33	+0.15	+0.03
FS10 Betting against beta	+0.25	+0.36	+0.78	+0.10	+0.31	+0.44	FS10 Betting against beta	+0.09	+0.16	+0.33	+0.04	+0.16	+0.15
FS11 Residual momentum	+0.28	+0.23	+0.15	+0.42	+0.26	+0.20	FS11 Residual momentum	+0.15	+0.20	+0.09	+0.11	+0.14	+0.12
FS12 Buy at the 52-week low	-0.62	-0.96	-0.64	-0.51	-0.41	-0.78	FS12 Buy at the 52-week low	-0.37	-0.64	-0.24	-0.23	-0.38	-0.38



2. What the twelve factsheets say together

1. **Two roots carry almost everything.** The momentum family (FS03, FS09, FS11, and the breakout inside FS01) and the low-risk family (FS02, FS04, FS10) are where the beta-adjusted alphas are. Within each family the signals are close substitutes.
2. **Raw long/short numbers mislead in a bull market.** The long/short books of FS03, FS04, FS09 carry market betas of -1.09 to -0.28 against the equal-weight universe; in a rising market that short beta alone cost 3–13% a year (beta × universe return). Beta-neutral legs remove most of this drag, but the neutralised books rarely pass the gate.
3. **Costs matter most for the fast signals.** Trading costs take 3.9–4.9% a year from short-term reversal (FS08) and 4.0–5.0% from seasonality (FS06), against 1.7–2.5% for 12-1 momentum and 1.0–1.6% for low volatility; both fast signals are negative after costs in most markets.
4. **Folklore fails where it contradicts momentum.** Buying at the 52-week low (FS12) and buying breakouts with tight stops on single stocks (FS01) both lose; the monkeys (FS05) only reflect whether small beat big.
5. **Survivorship was worth 2–5 percentage points a year** on the equal-weight universes outside the US; moving to point-in-time membership changed several conclusions (FS02).
6. **Six markets are not six tests.** The long/short books co-move strongly across universes (World contains the US; EU contains the Nordic blue chips), so the six markets are worth only 1.4–1.8 independent tests (effective number = $36 / \text{sum of the } 6 \times 6 \text{ correlation matrix}$). "Positive in all six" is weaker evidence than it sounds.

Strategy (L/S net)	Mean pairwise correlation	US–World	EU–SCANDI	Effective number of markets
FS03 Momentum	0.55	0.89	0.59	1.6
FS04 Low volatility	0.61	0.92	0.61	1.5
FS09 52-week high	0.63	0.92	0.61	1.4
FS10 Betting against beta	0.59	0.90	0.72	1.5
FS11 Residual momentum	0.47	0.84	0.52	1.8
FS12 Buy at the 52-week low	0.48	0.85	0.63	1.8

1. **Across the whole programme, few winners survive.** The factsheet trial ledger holds 682 gated tests. Under a programme-wide Benjamini–Hochberg correction 53 positive results survive (21 also Bonferroni, $|t| > 3.97$) against 21 reliable losers; the best positive candidates have a deflated Sharpe ratio of 0.68 at most even on the lenient bound, below the usual 0.95 (planning/FACTSHEET_TRIAL_LEDGER.md).

3. Gap analysis

Factor coverage today: stock-level in the six universes

Momentum & trend	covered	covered	covered	covered	covered	covered
Short-term reversal	covered	covered	covered	covered	covered	covered
Low risk (vol, beta, MAX)	covered	covered	covered	covered	covered	covered
Seasonality	covered	covered	covered	covered	covered	covered
Size	covered	partial	partial	partial	partial	partial
Value	partial	not covered	partial	partial	not covered	partial
Quality / profitability	partial	not covered	partial	partial	not covered	partial
Investment / asset growth	partial	not covered	partial	partial	not covered	partial
Earnings momentum / PEAD	partial	not covered	partial	partial	not covered	partial
Analyst revisions	not covered	not covered	not covered	not covered	not covered	not covered
Dividend / payout	partial	not covered	partial	partial	not covered	partial
Liquidity	partial	partial	partial	partial	partial	partial
Short interest	not covered	not covered	not covered	not covered	not covered	not covered
News / sentiment	not covered	not covered	not covered	not covered	not covered	not covered
Option-implied risk	not covered	not covered	not covered	not covered	not covered	not covered
	US	EU	UK	DK	SCANDI	World

partial = stock level in the US only (Sharadar fundamentals, not yet built) or factor-level returns elsewhere (JKP: UK, DK, World; none for EU, SCANDI); liquidity = traded-value proxy only

3.0 Closed since the first edition (27 Sep 2026)

Category	Gap	What was done
Data	No company fundamentals outside the US	Partly closed. US stock level (FS13b); outside the US at factor level (FS17): the fundamental pair (profit growth + issuance) adds to low beta and momentum in the UK, Denmark and Europe in both 2013–25 (JKP) and 1999–2013 (ART). Stock-level non-US history still needs a data source.
Data	No FX conversion; World sums local-currency returns	World converted to USD, unhedged, with Fed H.10 daily rates (EUR, GBP, DKK, SEK) and Yahoo PLN from 2015; the 21 Polish names stay in PLN before 2015 (<code>data.to_usd</code>).
Model	Programme-level multiple testing not yet consolidated	One ledger of every gated test in FS01–FS13 with programme-wide Bonferroni, Benjamini–Hochberg and deflated Sharpe (<code>code/ledger.py</code> , <code>planning/FACTSHEET_TRIAL_LEDGER.md</code>).
Implementation	Flat 10 bp per side; no size- or liquidity-dependent costs, no market impact	FS16: half-spread from traded value plus square-root market impact, at \$1m to \$10bn per book; the shortlist holds a positive Sharpe ratio to about \$1bn in the US, \$130m in World and \$13–29m in the EU, UK and Denmark (<code>code/impact.py</code>).

Category	Gap	What was done
Implementation	Short side: no borrow fees or availability in the sorts (50 bp flat in FS01 only)	Every short leg pays a size-tiered borrow fee: 0.25% a year for the largest half of the universe, 0.75% for the next 30%, 2% for the smallest 20% (<code>code/borrow.py</code>); FS01 keeps its flat 0.5%.
Implementation	Leverage without financing: BAB ($r_f = 0$, no spread), vol targeting up to 3×	Net long cash in beta-neutral and BAB books is charged at the USD risk-free rate + 0.5% (<code>borrow.financing</code>). Volatility targeting scales a dollar-neutral book and is not charged for cash; its margin cost is still missing.
Validation	No replication check against published factor returns	FS13: our US signals correlate 0.8 or more with the matching JKP factor for 8 of 12.
Validation	No sector or industry neutrality	FS19 (pre-registered, passed): ranked within 11 sectors the shortlist keeps +5.3% a year (t 3.0), two thirds of its unrestricted return; UK sector coverage only 34%.
Validation	No stress tests of named episodes (2015–16, COVID crash and rebound, 2022 rate shock) and no crash-risk analysis for momentum	FS19: five episodes 2015–2022 for market, low beta, momentum and the FS18 overlay; the overlay stays within $\pm 8\%$ in every episode; low beta lost 24–29% in the US and World in the COVID crash and in 2022. Momentum-crash regression: sign as in Daniel & Moskowitz before 2013 (t –1.9), untestable after (too few bear months).
Validation	No unit tests of the engine (look-ahead, signal lags, cost accounting)	FS19: signals identical when rebuilt from truncated data (171 comparisons), iid-noise t -statistics behave as expected (4.2% beyond 1.96), a next-month cheat signal is caught ($t > 180$), cost arithmetic exact. Flag: 'beta-neutral' low beta keeps a realised beta of +0.3 to +0.6 in the US, UK, EU and World.
Process	Factsheet trials outside the programme ledger	All factsheet trials are now in the factsheet trial ledger; FS14 will be pre-registered there.

3.1 Priority gaps still open (impact high, effort lowest first)

Category	Gap	Impact	Effort	Remedy
Validation	The six universes overlap and co-move: World contains US, UK and EU; EU contains the Danish, Swedish and Finnish blue chips	High	Low	Report the effective number of markets; judge on pooled evidence; treat World as a summary, not a seventh market; run EU ex-Nordics
Data	Residual survivorship outside the US: 9–25% of index member-quarters have no price (mostly delisted names)	High	Medium	Buy delisted-inclusive history for Europe, or lengthen with the ART archive (1996–2013)
Model	Sort-based tests only; no multivariate (Fama-MacBeth) regressions	High	Medium	Monthly cross-sectional regressions on all library signals, per universe
Coverage	Missing factor families: value, quality, investment, earnings momentum, analyst revisions, payout, liquidity	High	Medium (US) / High (non-US)	US stock-level from Sharadar now; JKP factor-level for UK, DK, World
Data	Oslo missing from SCANDI; DK has ~19 names	Medium	Low	Add OBX history; report DK only as a robustness market
Model	Raw 252-day betas, no shrinkage	Medium	Low	Vasicek or Frazzini-Pedersen shrinkage
Model	No factor model for Europe or SCANDI inside JKP; French Europe used, in USD	Medium	Low	Build local-currency factor returns from the library itself
Model	Equal weighting and one-month holding everywhere	Medium	Low	Add value-weighted (US) and overlapping 3/6/12-month holdings

3.2 Data

Gap	Where it bites	Impact	Effort	Remedy	Source we already have
Residual survivorship outside the US: 9–25% of index member-quarters have no price (mostly delisted names)	all non-US results; short legs penalised, long legs flattered	High	Medium	Buy delisted-inclusive history for Europe, or lengthen with the ART archive (1996–2013)	ART PIT database (Project1, to 2013); none for 2013–26
No shares outstanding outside the US	FS07 size and the FS05 index benchmark use traded value as a proxy	Medium	Medium	Shares-outstanding history or free-float index weights	none
Short history (from 2012)	FS06 seasonality averages 1–13 years (paper: 20); FS11 starts 2014; holdout only 6.7 years	Medium	Medium	Extend back with ART (1996–2013) for EU, UK, DK, US	ART PIT database
Oslo missing from SCANDI; DK has ~19 names	SCANDI and DK sorts use 3–5 groups of 6–12 stocks; noisy	Medium	Low	Add OBX history; report DK only as a robustness market	none
US prices without high/low or volume in our Sharadar extract	FS01 true-range N; US size-independent liquidity signals	Low	Low	Re-pull Sharadar SEP with OHLCV	Sharadar subscription
Event and positioning data not used: earnings dates, analyst revisions, short interest, options, news	360° gaps in FS02 (news vs no-news jumps, implied skew, borrow)	Medium	Medium–High	Check coverage of the Project2 caches, start with the US	Project2 pit_short_positions, pit_options, pit_headlines, pit_13f; ART IBES (to 2013)

3.3 Model

Gap	Where it bites	Impact	Effort	Remedy	Source we already have
Sort-based tests only; no multivariate (Fama-MacBeth) regressions	overlapping signals (momentum family, low-risk family) are judged one at a time	High	Medium	Monthly cross-sectional regressions on all library signals, per universe	code/signals*.py library
Raw 252-day betas, no shrinkage	FS10 BAB, beta-neutral fixes, attribution	Medium	Low	Vasicek or Frazzini-Pedersen shrinkage	—
Residual momentum on a market-only model, 24 months	FS11	Low	Low	Fama-French 3-factor residuals (US), 36 months when history allows	French factors (Project2 cache)
No factor model for Europe or SCANDI inside JKP; French Europe used, in USD	attribution alphas for EU and SCANDI	Medium	Low	Build local-currency factor returns from the library itself	library L/S returns
Equal weighting and one-month holding everywhere	turnover-heavy strategies (FS08, FS06); comparability with papers that value-weight or hold 6–12 months	Medium	Low	Add value-weighted (US) and overlapping 3/6/12-month holdings	—

3.4 Implementation

Gap	Where it bites	Impact	Effort	Remedy	Source we already have
No taxes, dividend withholding or currency hedging costs	Danish and cross-border investors	Medium	Medium	Investor-specific net-of-tax layer	Project1 art_measured_taxes
Close-to-close execution, monthly rebalance at month-end	all strategies; month-end crowding	Low	Low	Next-day VWAP proxy; staggered rebalance days	—

3.5 Coverage

Gap	Where it bites	Impact	Effort	Remedy	Source we already have
Missing factor families: value, quality, investment, earnings momentum, analyst revisions, payout, liquidity	the multifactor battery would be price-only	High	Medium (US) / High (non-US)	US stock-level from Sharadar now; JKP factor-level for UK, DK, World	Sharadar US fundamentals; JKP

3.6 Validation

Gap	Where it bites	Impact	Effort	Remedy	Source we already have
The six universes overlap and co-move: World contains US, UK and EU; EU contains the Danish, Swedish and Finnish blue chips	every 'positive in k of six' count; six markets are worth fewer than two independent tests	High	Low	Report the effective number of markets; judge on pooled evidence; treat World as a summary, not a seventh market; run EU ex-Nordics	the L/S series already on disk
No capacity estimate	which books survive at realistic AUM, above all DK and SCANDI	Medium	Medium	Capacity at a participation cap (e.g. 5% of daily traded value) per book	traded-value panels (PIT)

3.7 Process

Gap	Where it bites	Impact	Effort	Remedy	Source we already have
Repo assembly script does not know the factsheets	publishing	Low	Low	Add factsheets to assemble_repo.py	—

4. Where the series ended up (FS13–FS19)

Steps 1–5 of the first edition's plan are done, including the remaining cheap gaps (FS16 costs, FS19 engine tests, stress episodes and sector neutrality). In short:

Step	Result	Factsheet
Signal battery across all six markets	Low beta (beta-neutral) is positive in 6 of 6 markets and 12-1 momentum in 5 of 6; no price signal survives the correction on its own	FS13

Step	Result	Factsheet
US fundamentals, stock level	Profit growth and debt issuance add alpha; value and investment do not	FS13b
Multivariate test	Residual momentum and 52-week high are redundant next to 12-1 momentum	FS13c
Pre-registered multifactor model	No selection rule beats equal weights (t 1.4)	FS14
ART history, 1999–2013	The shortlist holds out of sample (Sharpe 0.8); selection rules fail again	FS15
Costs that grow with size	Positive to about \$1bn in the US, tens of millions in Europe and the UK, never in the Nordics	FS16
Fundamentals outside the US	Profit growth + issuance add to the shortlist at factor level (t 4.7)	FS17
The portfolio I would run	Market + 5%-vol overlay: Sharpe +0.15–0.21 in the US, EU, UK and World at \$50m per book; pooled t 1.85, not passed	FS18
Robustness	Engine clean; within-sector shortlist keeps two thirds (t 3.0); "beta-neutral" low beta is not market-neutral in crashes	FS19

Still open: stock-level fundamentals outside the US (data request in `planning/DATA_REQUEST_LETTER.md`); realised-beta control for the low-beta book (shrunk betas, as in Frazzini & Pedersen); cost-model calibration to real spreads; UK sector labels for investment trusts.

5. Reproduce

`code/build_overview.py` reads `results/xs_*.json`, `turtle_summary.json`, `lottery_summary.json`, `fix_summary.json`, `monkey_summary.json` and the neighbourhood files.